

Avionics

This subteam is the most important and critical for the whole team, as it is responsible for breathing life into a UAV built by the fabrication team and designed by the design team, allowing it to actually take to the sky. To become proficient in avionics, you need to read a lot, particularly the ArduPilot documentation, which is the autopilot firmware we use.

1. Copter Basics

Get a basic understanding of what a copter is, and how does arducopter works. You can start by reading the following sections, and then optionally move to read the other sections as well of <https://ardupilot.org/copter>

- [Introduction](#)
- [How copter works](#)
- [Flight features](#)
- [Copter Safety](#)
- [First time Setup](#) (important)
- [First time Flying](#) (important)

Optionally, you can also repeat the process for [Arduplane](#) as well.

2. Mission Planner

This is the primary Ground Control Station that we use is Mission Planner. To play around with its features, you can connect with a Simulator like [this](#). After running it, read the following articles to get the understanding of the software in a bit more depth. Optionally you can read the complete [documentation](#) for mission planner as well.

- [Installation](#) (if you haven't installed it)
- [Mission planner screens](#)
- [Mission Planning](#)

Task: Create a flight plan for a quadcopter simulation starting from the GIKI Football Ground till library. Add a minimum of 7 waypoints with at least 3 different altitude levels. Ensure the drone returns to launch (RTL) at the end of the mission.

3. Flight Time Calculations

Now let's go back a bit. Before a UAV is even conceptualized by the design team, it is the responsibility of the avionics team to provide a list of electronics, particularly the propulsion system (motors and battery). One of the constraints given by the competition is the Maximum Takeoff Mass (MTOM), along with the maximum flight time. The avionics team shortlists a set

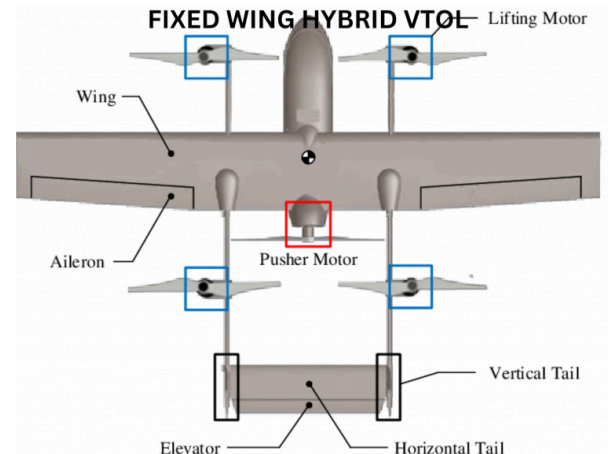
of motors that can provide the required thrust for the given MTOM, performs calculations to determine the battery size that can supply the required current for the required duration, and selects the system that is most efficient, that is, the one that provides the necessary thrust with the least weight. Here is an [example](#)

Task: Select the model for lift motors as well as a push motor for a 4+1 vtol with 7.5 kg MTOM and 10 minute maximum flight time. Select at least 3 models for lifting motors and 3 for pusher motor. Use the following thrust to weight ratios (T/W):

- Takeoff for 10 seconds (lift motors) = 2:1
- Land for 10 seconds (lift motors) = 0.6:1
- Cruise for 1 minute (lift motors) = 1.2:1
- Cruise for 8 minutes 40 sec (pusher motor) = 0.9:1

Use [tmotors](#) to search for motors. Use the specification tab of each motor to determine the thrust and current required at the thrust level. If the required thrust is not mentioned in the table, use the following interpolation formula between two rows a and b to get the required current

$$\text{current required} = \text{current } a + (\text{thrust required} - \text{thrust } a) \frac{\text{current } b - \text{current } a}{\text{thrust } b - \text{thrust } a}$$



Test Data										Export to Excel
Type	Propeller	Throttle (%)	Voltage (V)	Current (A)	Power (W)	RPM	Torque (N-m)	Thrust (g)	Efficiency (g/W)	Operating Temperature (°C)
6007II-X KV160(Alpha 60A 12S)	MFZ2211	40%	47.92	3.65	175	2642	0.52	1926	11.02	64 (Ambient Temperature : 25)
		45%	47.89	4.89	234	2926	0.63	2362	10.09	
		50%	47.85	6.40	306	3201	0.75	2831	9.25	
		55%	47.81	8.13	389	3467	0.88	3322	8.55	
		60%	47.75	10.17	485	3730	1.01	3846	7.92	
		65%	47.70	12.39	591	3975	1.15	4369	7.39	
		70%	47.64	14.84	707	4206	1.29	4895	6.92	
		75%	47.57	17.51	833	4421	1.42	5419	6.50	
		80%	47.50	20.34	966	4624	1.55	5926	6.13	
		85%	47.45	22.69	1076	4771	1.65	6291	5.84	
		90%	47.44	22.79	1081	4768	1.65	6299	5.83	
		95%	47.44	22.85	1084	4760	1.65	6297	5.81	
		100%	47.44	22.91	1087	4754	1.65	6291	5.79	
		40%	23.90	5.14	123	2324	0.40	1486	12.08	

Example Data

SDES

SDES stands for Software Development and Embedded Systems. The competitions usually require performing computer vision tasks such as object detection and classification, followed by actions based on the results of these tasks, such as dropping a payload on a target object. Since this functionality is not available in the standard ArduPilot firmware, and flight controllers are not capable of handling computer vision models, the SDES team is responsible for training these models, developing and testing Python scripts related to mission objectives, and deploying them on a companion computer, such as the Jetson Nano. The team is also responsible for additional tasks, such as developing Team Foxtrot's website.

1. Basics

This session introduced the syntax of Python and some basic data structures, such as lists and dictionaries, before progressing to solving CS101-style problems. You cannot begin working on scripts without knowing how to solve basic problems, that is, how to break a problem down into individual components, develop solutions for each component, test and debug each one, combine the components, and then debug any issues related to their integration. To work effectively on SDES projects, you need to strengthen your problem-solving and debugging skills, as libraries and even different programming languages can be learned relatively easily once these skills are developed.

[Here](#) is the resource for the basics syntax. Follow and understand all the pages under the “Python Tutorial” section, especially [List](#), [Tuples](#), [Sets](#), [Dictionaries](#), [For loops](#), [Functions](#), [Range](#), [Arrays](#), and [String formatting](#). Just reading is not enough, as you will forget the material shortly afterward. You should use the “Try It Yourself” feature on W3Schools and experiment with the given code to generate different outputs. Your goal should be to clearly understand these language tools so that you can use them effectively when solving basic problems.

Task: Solve [these](#) questions without looking at the solution or using any LLM or AI agent, even if it takes you hours. The goal is to develop your problem-solving skills, and copy-pasting or vibe coding solutions will not help you achieve that. Use the resources above for reference. If you are struggling, try solving the problem conceptually on paper before translating it into code. If you encounter a syntax error, carefully read and understand the error message, then fix it. If you are not getting the desired output, revisit your planned solution and identify what you may have missed. If the solution works on paper, focus on the implementation details by printing variables at each step to verify their values, this will help you pinpoint where the issue occurs.

2. Object Oriented Programming

This is an extension of the basics. When your codebase starts to grow beyond a couple of files, Object oriented programming can help you organize it more cleanly. Follow [this tutorial](#) from w3schools or/and [this tutorial](#) on github.

Task 1: You are designing a small employee–management system.

1. Create a base class Person with

- private attributes `__name` and `__age`
- a constructor
- getter and setter methods with validation (age must be > 0)

2. Create a derived class Employee(Person) that adds:

- attributes: `employee_id`, `salary`
- overrides the parent constructor
- uses `super()` to initialize the Person attributes
- makes salary a protected attribute (`_salary`)
- method `increment_salary(amount)` that increases salary only if `amount > 0`

3. Create another derived class Manager(Employee) that adds:

- a list of `team_members`
- method `add_member(employee)`
- method `show_team()`
- Manager should override `increment_salary` to add a 10% bonus on top of the increment.

Task 2: You must design a system to manage bank accounts.

1. Create a class Account with:

- attributes: `owner`, `balance`
- a constructor
- a static method `validate_amount(amount)` that returns True if `amount > 0`
- a class method `from_string(data_str)` that creates an Account from: "owner,balance"

(This tests the factory method pattern.)

2. Create two subclasses:

- 2.1 SavingsAccount(Account)
- 2.2 CurrentAccount(Account)

Each should override a method `account_type()` returning a string.

3. Create a class AccountManager that stores multiple accounts and has:

- method `add_account(account)`
- method `transfer(acc1, acc2, amount)` that:

- uses the static method to validate amount
- transfers money only if acc1 has enough balance

Note: Same rules described in the previous task applies to this. That is, no use of LLMs or assistants.

3. Using Git and Github

Github is fundamental for version control and collaboration. Follow these slides to get a basic idea. You can also watch these tutorials.

▶ Git Will Finally Make Sense After This

▶ Master GitHub: From Beginner to Expert in 46 min

Task: For this task, you will need to pair up with another member. If you are unable to do so, you can clone the repository twice with different names, and assume the role of both Person A and Person B. Using LLM or assistants is permitted for this assignment, but with a condition. Don't just copy paste commands. Try to read and understand the solution. Furthermore, read the documentation of the commands you are using to get a better understanding.

☰ Activity Git & GitHub LnD

4. Using Terminal

Knowing your way around the terminal is one of the most useful skills that you will appreciate you investing your time into. Although we didn't get a chance to cover it this semester, still go through [this article](#) in order to understand basic terminal commands, which will make your life easier when working with real-world projects. It is recommended that you [install WSL](#) (Windows Subsystem for Linux) and follow along with the tutorial.

Design

1. Aircraft Design

Task: You are required to **design a fixed-wing aircraft** with a **maximum takeoff weight (MTOW) of 2 kg**, subject to the aerodynamic and geometric constraints listed below.

Your design should clearly justify all assumptions, calculations, and final geometric parameters derived from the given data.

Aircraft Requirements

- Maximum Takeoff Weight (MTOW): **2 kg**
- Maximum Lift Coefficient (C_L max): **1.6**
- Stall Velocity, (V_s): **12 m/s**
- Wing Aspect Ratio (AR): **6.5**
- Wing Taper Ratio: **0.7**
- Slenderness Ratio: **6**

Horizontal Tail Parameters

- Horizontal Tail Volume Coefficient, (VHT): **0.5**
- Aspect Ratio of Horizontal Tail: **2**
- Taper Ratio of Horizontal Tail: **1**

Vertical Tail Parameters

- Vertical Tail Volume Coefficient, (VVT): **0.04**
- Aspect Ratio of Vertical Tail: **2**
- Taper Ratio of Vertical Tail: **0.5**

Documentation

1. Formatting for Critical Design Review

Documentation plays a central and mission-critical role within Team Foxtrot, serving as the primary medium through which the team communicates its engineering intent, design rationale, and operational readiness to competition authorities and external evaluators. While the aircraft itself represents the physical outcome of the project, the documentation reflects the intellectual rigor, systematic engineering approach, and compliance with competition standards that underpin the entire development process. It ensures that every design decision is transparent, traceable, and defensible.

The CDR represents a more advanced and rigorous phase, where the finalized design is presented in detail. It documents the complete design process, manufacturing methodology, system integration, testing procedures, and performance validation. Special emphasis is placed on flight safety, reliability, and operational capability, including risk assessments, redundancy planning, and compliance with safety regulations. Through the CDR,

Task: CDR Write-Up

All members assigned sections of the Critical Design Review (CDR) must complete their write-ups strictly according to the defined standards. Content should clearly explain the finalized design, justification, implementation approach, and how the subsystem meets performance and safety requirements.

Formatting requirements:

- Font: Times New Roman
- Body text: 12 pt
- Headings: 13 pt

This document is intended for a technical audience, so appropriate engineering jargon and technical detail may be used. Ensure clarity, accuracy, and consistency so that your section can be directly integrated into the final CDR without revisions

Media

This subteam is responsible for shaping the public image of the team and communicating our work effectively to both internal and external audiences. The media team documents progress, manages social media presence, designs visual content, and ensures that our achievements, events, and technical milestones are presented in a clear, engaging, and professional manner. A strong media presence helps attract sponsors, inspire audiences, and showcase the hard work of all subteams.

To become proficient in the media team, you should have a basic understanding of content creation, visual storytelling, and digital platforms. Familiarity with tools like Canva, Photoshop, Filmora, or other basic software is a plus, along with an eye for detail and creativity.

1. Basics

Develop a foundational understanding of how this subteam supports the technical teams. This includes learning how to capture high-quality visuals, design clear and consistent graphics, and produce short-form videos suited for digital platforms. Emphasis should be placed on clarity, consistency, and effective communication rather than just aesthetics.

Photography focuses on documenting the team's activities, and the UAV itself. The goal is to capture clear, well-composed images that tell a story and follow basic photography principles such as framing, lighting, and focus.

Task: Choose one common household object (e.g., cup, notebook, headphones, plant, watch, bottle) and capture 3 photographs of the same object with the following requirements:

- At least 1 wide shot
- At least 2 close-up/detailed shots

The images should demonstrate:

- Proper focus and exposure
- Thoughtful framing (rule of thirds encouraged)
- Consistent visual style

No heavy editing is required; basic color correction is sufficient.

Videography involves creating short and engaging videos to showcase the team's progress, behind-the-scenes moments, and highlights. This includes planning shots, recording stable footage, and editing clips to create videos suitable for social media platforms.

Task: Using the same object chosen for photography, create a 20–30 second Instagram Reel.

Requirements:

- Multiple short clips (2–4 seconds each)
- Variation in angles and shot sizes
- Clean cuts or simple transitions
- Background music (optional but encouraged)
- The reel should feel visually cohesive and intentionally paced.

Graphic design is responsible for creating posters, announcements, and digital creatives that align with the team’s branding. This domain focuses on layout, color harmony, typography, and visual hierarchy to communicate information clearly.

Task: Using the same everyday object chosen for photography and videography, design a product specification poster as if the object were being marketed online.

Requirements:

- Product name (you may invent one)
- At least 4 clear specifications (e.g., material, size, capacity, weight, feature)
- One clear product visual (photo or illustration)
- Clean layout with proper visual hierarchy
- Limited color palette (2–3 colors recommended)
- Readable typography

The poster should feel informative, minimal, and well-structured rather than decorative.

2. Content & Outreach

This section focuses on how media is used strategically to engage audiences and maintain a strong online presence. You should understand how to plan, schedule, and publish content in a way that aligns with platform algorithms and audience behavior, particularly on Instagram.

Task: Create a strategic Instagram content plan for one month for the team. The plan should include:

- Posting frequency and content types (posts, reels, stories)
- Proposed posting days and times
- Clear objectives for each post (awareness, recruitment, documentation, engagement)
- Target audience for the content

- Brief description of each post's idea or theme
- Consistency in branding and tone.

You could display this in the form of a table or a document.

Submit all the tasks [here](#)